

Pigeonhole Principle

IMO Training Level 2

17/07/2021

This notes is intended to be a Combinatorics notes, but since Pigeonhole principle is so extensively used, there are also plenty of Algebra and Number Theory problems.

Basic Questions

1. Given 9 points inside a square of side length 1 in general position (i.e. no 3 points collinear), show that we can always choose 3 points which form a triangle with area not exceeding $\frac{1}{8}$
2. 89 points lie inside a circle of radius 5, show that there are two points, the distance between which is less than $\sqrt{2}$
3. (JBMO 1999 P3) Let S be a square with side length 20 and let M be a set of points containing the vertices of S plus another 1999 points lying inside S . Prove that there exists a triangle with vertices in M and with area at most $\frac{1}{10}$. (here we consider degenerate triangle as a triangle with area 0)
4. Let n be an odd positive integer not divisible by 5. Show that there exists a multiple of n whose every digit (in decimal base) is 1.
5. Let $p > 3$ be a prime number. Show that there are positive integers x, y, k with $0 < 2k < p$ such that

$$kp + 3 = x^2 + y^2$$

Intermediate Questions

6. (APMO 2008 P2) Students in a class form groups, each of which contains exactly three members, such that any two distinct groups have at most one member in common. Prove that, when the class size is 46, there is a set of 10 students in which no group is properly contained.
7. (USAJMO 2010 P5) Two permutations $a_1, a_2, \dots, a_{2010}$ and $b_1, b_2, \dots, b_{2010}$ of the numbers $1, 2, \dots, 2010$ are said to *intersect* if $a_k = b_k$ for some value of k in the range $1 \leq k \leq 2010$. Show that there exist 1006 permutations of the numbers $1, 2, \dots, 2010$ such that any other such permutation is guaranteed to intersect at least one of these 1006 permutations.
8. (USAMO 2009 P2) Let n be a positive integer. Determine the size of the largest subset of $\{-n, -n+1, \dots, n-1, n\}$ which does not contain three elements a, b, c (not necessarily distinct) satisfying $a + b + c = 0$.
9. Let a be a fixed natural number. Prove that the set of prime divisors of $2^{2^n} + a$ (where n runs over $1, 2, \dots$) is infinite
10. (IMO 2016 A2) Find the smallest real constant C such that for any positive real numbers a_1, a_2, a_3, a_4, a_5 (not necessarily distinct), one can always choose distinct subscript i, j, k, l such that

$$\left| \frac{a_i}{a_j} - \frac{a_k}{a_l} \right| \leq C$$

Hard Questions

11. (IMO 2017 N3) Determine all integers $n \geq 2$ with the following property: for integers $a_1, a_2, \dots, a_n$ whose sum is not divisible by n , there exists an index $1 \leq i \leq n$ such that none of the numbers

$$a_i, a_i + a_{i+1}, \dots, a_i + a_{i+1} + \dots + a_{i+n-1}$$

is divisible by n . (We let $a_i = a_{i-n}$ when $i > n$)

12. (IMO 2010 C2) On some planet, there are 2^N countries ($N \geq 4$) Each country has a flag N units wide and 1 unit high, composed of N fields of size 1×1 , each field being either yellow or blue. None of the two countries have the same flag.

We say that a set of N flags is *diverse* if these flags can be arranged into an $N \times N$ square such that all N fields on its main diagonal will have the same color. Determine the smallest positive integer M such that among any M distinct flags, there exist N flags forming a diverse set.

13. (IMO 2011 N2) Consider a polynomial $P(x) = (x + d_1)(x + d_2) \dots (x + d_9)$, where $d_1, d_2, \dots, d_9$ are nine distinct integers. Prove that there exists an integer N such that for all integers $x \geq N$, the number $P(x)$ is divisible by a prime number greater than 20.
14. (IMO 2010 C3) 2500 chess kings have to be placed on a 100×100 chessboard so that:
- (i) no king can capture any other one (i.e. no two kings are placed in two squares sharing a common vertex);
 - (ii) each row and each column contains exactly 25 kings.

Find the number of such arrangements. (Two arrangements differing by rotation or symmetry are supposed to be different.)

15. (Polish MO 2009 Round 3 P2) Let S be a set of all points of a plane whose coordinates are integers. Find the smallest positive integer k for which there exists a 60-element subset of set S with the following condition satisfied:

for any two elements A, B of the subset there exists a point C contained in S such that the area of triangle ABC is equal to k .